

Beginning Algebra Test III

Name
Date
Course

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the number, and show work when necessary.
- iii) Draw a rectangle around your answers.
- iv) Staple a copy of these test questions with your work and your final answers.

Solve.

- ① $-6 = 4 - 2\sqrt{x}$
- ② $5y^3 + 20y^2 = 2y + 8$
- ③ $3x^2 + 9x = 0$
- ④ $8x^2 = 4 - 3|x|$
- ⑤ $\sqrt{y} + 5 = 9$
- ⑥ $\frac{\sqrt{2b-3}}{4} - 8 = -6$
- ⑦ $\sqrt{2x^2 - 15} = \sqrt{-x}$
- ⑧ $2y^2 - 7 = -5$
- ⑨ $4(x-1)^2 + 2 = 5$
- ⑩ $-8x^3 = -72x$
- ⑪ $-\frac{1}{3} + \frac{2x}{9} = 4x - 1$
- ⑫ $-80x^3 = -x^5 + 81x$
- ⑬ $3y^6 + 27y^5 = -60y^4$
- ⑭ $x - 6 = \frac{16}{x}$
- ⑮ $\frac{4}{x} - \frac{2}{5} = 1 + \frac{3}{5x}$
- ⑯ $\frac{1}{2y^2} = \frac{9}{2y^3} - \frac{9}{y^4}$
- ⑰ $x^3 - 2x^2 - x + 2 = 0$
- ⑱ $6x^3 - 2x^2 = -24x + 8$
- ⑲ $\frac{3(\frac{a}{2} - 7)^2}{5} + 6 = 10$
- ⑳ $-2 = (4x+1)^2 - 5$

$$\textcircled{21} \frac{1}{x-3} = \frac{4x+6}{x^2-5x+6} + \frac{5}{x-2}$$

$$\textcircled{22} \frac{1}{x-7} + \frac{1}{x} = \frac{7}{x^2-7x}$$

$$\textcircled{23} \frac{-20}{x^2-12x+32} = \frac{x+3}{x-4} - \frac{5}{x-8}$$

$$\textcircled{24} \sqrt{x+8} = 2 + \sqrt{x}$$

$$\textcircled{25} x+6 = \sqrt{25x+6}$$

Beginning Algebra Test III Answers

- ① $x = 25$
- ② $y = -4$, $y = \pm \frac{\sqrt{10}}{5}$
- ③ $x = 0$, $x = -3$
- ④ $x = -4$, $x = \frac{1}{8}$
- ⑤ $y = 16$
- ⑥ $b = 6\frac{1}{2}$ or $b = 33.5$
- ⑦ $x = -3$, $x \neq \frac{5}{2}$
- ⑧ $y = \pm 1$
- ⑨ $x = \pm \frac{\sqrt{3}}{2} + 1$
- ⑩ $x = 0$, $x = 3$, $x = -3$
- ⑪ $x = \frac{3}{17}$
- ⑫ $x = 0$, $x = 9$, $x = -9$
- ⑬ $y = 0$, $y = -4$, $y = -5$
- ⑭ $x = 8$, $x = -2$
- ⑮ $x = 1\frac{7}{7}$
- ⑯ $y = 3$, $y = 6$

- ⑰ $x = 1$, $x = -1$, $x = 2$
- ⑱ $x = \frac{1}{3}$, $x^2 \neq -4$
- ⑲ $a = \pm \frac{4\sqrt{15}}{3} + 14$
- ⑳ $x = \frac{\pm\sqrt{3} - 1}{4}$
- ㉑ $x = \frac{7}{8}$
- ㉒ No Solution, $x \neq 7$
- ㉓ $x = 2$, $x \neq 8$
- ㉔ $x = 1$
- ㉕ $x = 3$, $x = 10$